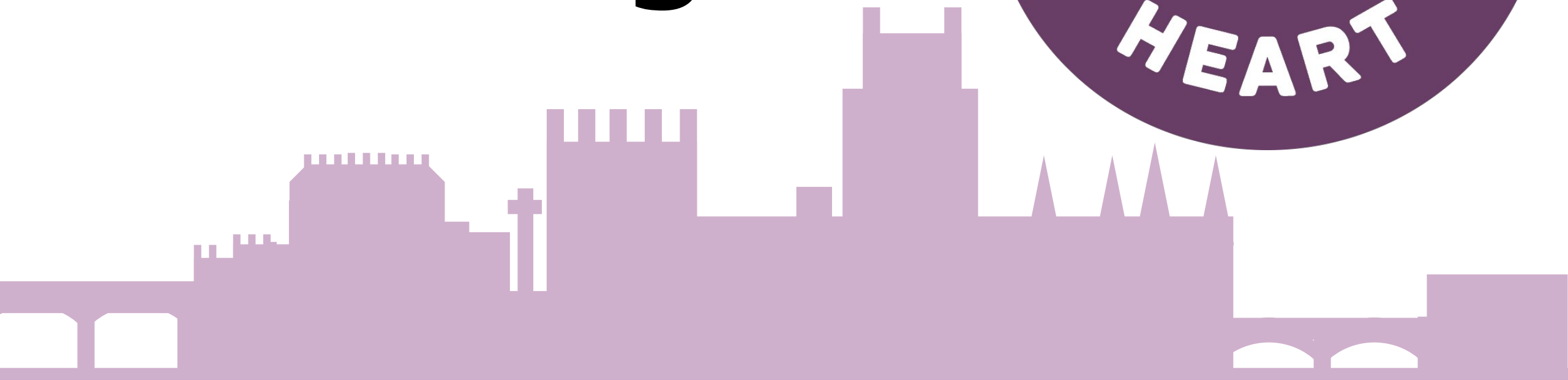


Electrical Meeting 1



@durham.heart



Durham Heart

Introduction



Progress Update

						Display Week:		1		Oct 13, 2025							Oct 20, 2025							Oct 27, 2025						Nov 3, 2025							
						13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	1	2	3	4	5	6							
TASK	ASSIGNED TO	PROGRESS	START	END		M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T
Improvement Integration																																					
Calculate the power loss of the current ltspice circuit on LTSpice, and implement improvements (eg. Replace componets, redesign)	Joe Gater, Samuel Finney	30%	10/15/25	10/29/25																																	
Develop and simulate gate-drive PWM control for the inverter using the ESP32 module in Wokwi (see connection on KiCAD Schematics)	Lavirotte Renan, Alex Johnson	100%	10/15/25	10/22/25																																	
KiCAD PCB routing learning	Lavirotte Renan, Alex Johnson, Anas Amraoui	0%	10/23/25	10/29/25																																	
Construct rectifier and boost converter KiCAD schematics	Joe Gater, Samuel Finney	0%	10/30/25	11/5/25																																	
Develop and simulate gate-drive PWM control for the rectifier in Wokwi.	Lavirotte Renan, Alex Johson	0%	10/30/25	11/5/25																																	
Frequency Tuning mechanism elucidation, method determination, and system development using LTSpice	Joe Gater, Samuel Finney	0%	10/15/25	11/5/25																																	
Simulate vertical, lateral, and radial coil misalignment effect on coupling, field, and losses using COMSOL, and determine the alignment tolerance	Ahaan Jaisinghani, Aoife McGuire	0%	10/15/25	11/5/25																																	
Coil testing design and conduction (max temperature to be measured); root cause analysis, and solution development; chracteristics extraction	Lim Zheng, Katrina Lai	0%	10/15/25	11/5/25																																	
Motor control invetigation, component selection and circuit construction for motor control in KiCAD Schematics	Meg Croft, Kian Nooralahiyan	0%	10/15/25	11/5/25																																	
Research into the importance, and integration of sesnors into the current block diagram: pressure senor, flow sensor, motor sensor, ECG sensor, etc	Sam Scandler, Clara Jang	0%	10/15/25	11/5/25																																	
Motor geometry optimisation using ANSYS	Scarlette, Rufus, Cody	0%	10/15/25	11/5/25																																	
External and internal battery hunting, and charge control component selection, and circuit design in KiCAD Schematics; calculate required external battery volatge through reverse engineering	Evie Smith, Habib Mosa	0%	10/15/25	11/5/25																																	

Social Event

**Thursday, October 30th
8pm
Head of Steam (top bar)**

Thank you for your hard work!

